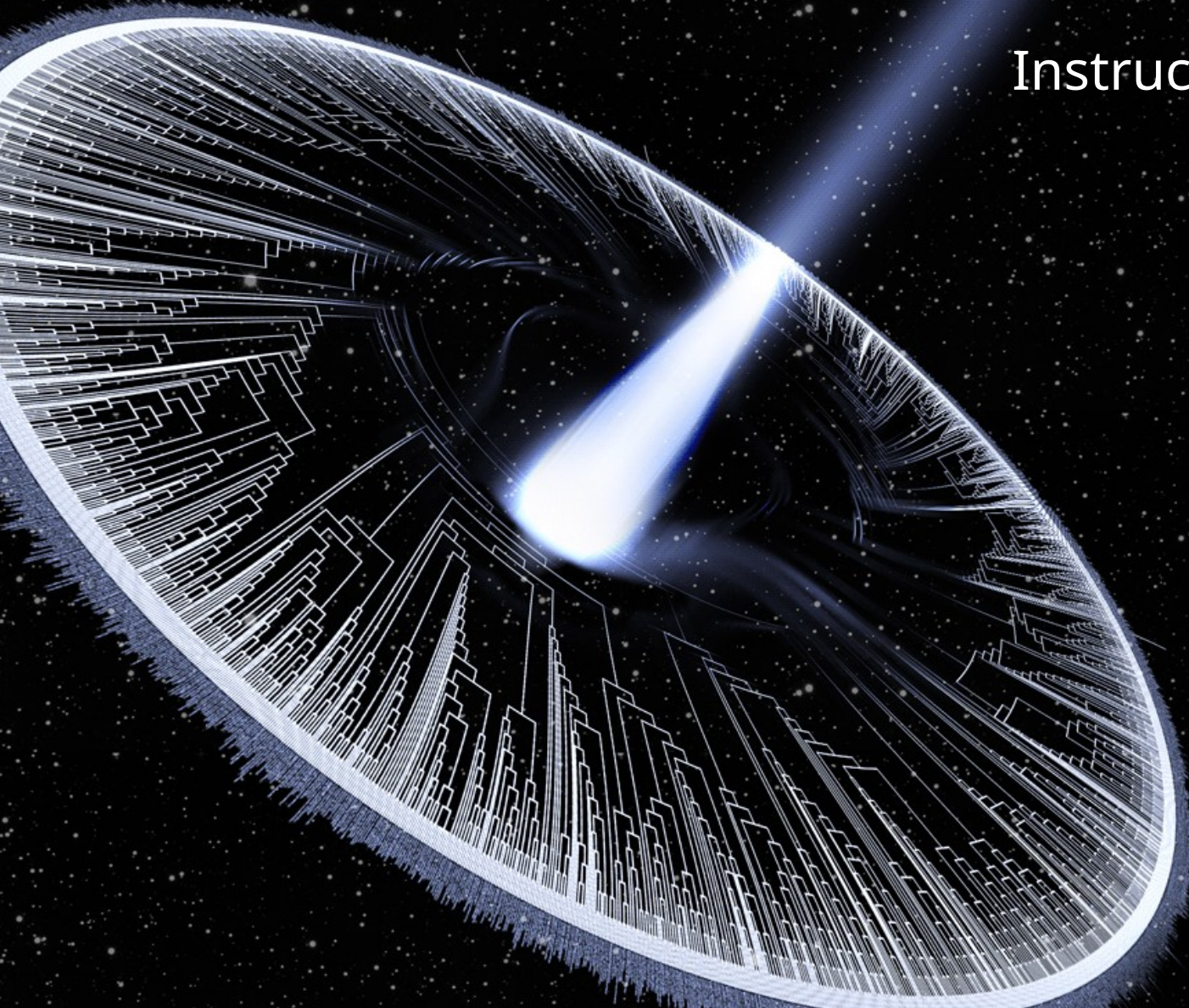


Macroevolution

Instructor: Josef Uyeda

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uyedalab.com



Goals of the course:

1. Learn the core of phylogenetic models
2. Learn how to use trees
3. Learn about the big questions in macroevolution
4. What are your goals?

Syllabus

- See Canvas Site
- (Syllabus subject to change, especially schedule)

Introductions

Your name, department, year, program

What topic you most want to learn about in this class

What's your best science "hot take"?

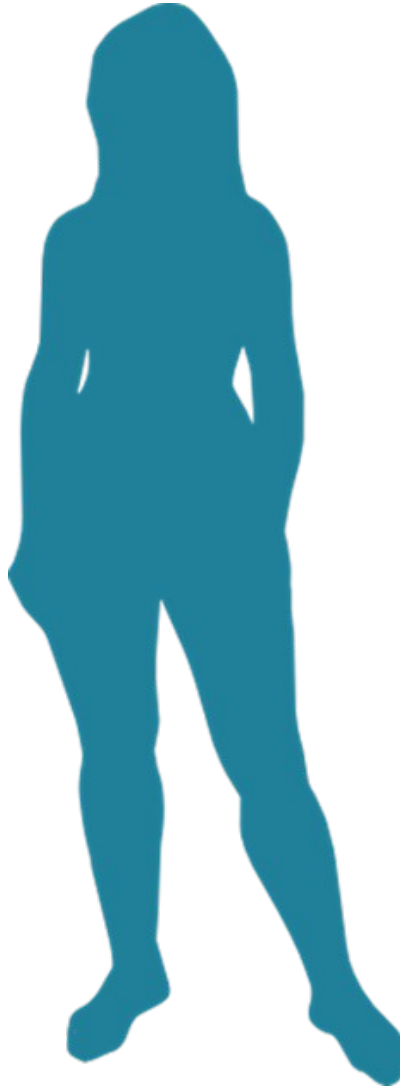
Why phylogeny?

- What does a phylogeny represent?
 - [OneZoom](#)
- We will use poll everywhere:
 - Permalink: www.pollev.com/josefuyeda941

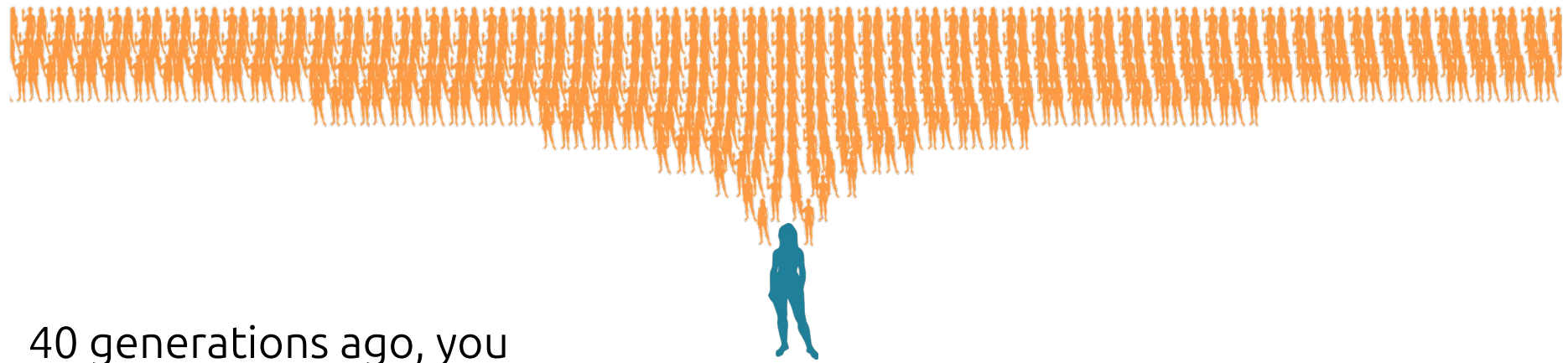
How many ancestors do you have?

- Consider your personal family tree (i.e. a pedigree). How many parents do you have? How many grandparents? Great grandparents? What's the formula?
- Approximately how many ancestors do we have 25 generations ago? (25 generations for humans is ~ 400-900 years)
- What are phylogenies? Are they a) hypotheses b) real facts that exist in nature or c) a metaphor?

Consider your personal family tree



Counting ancestors



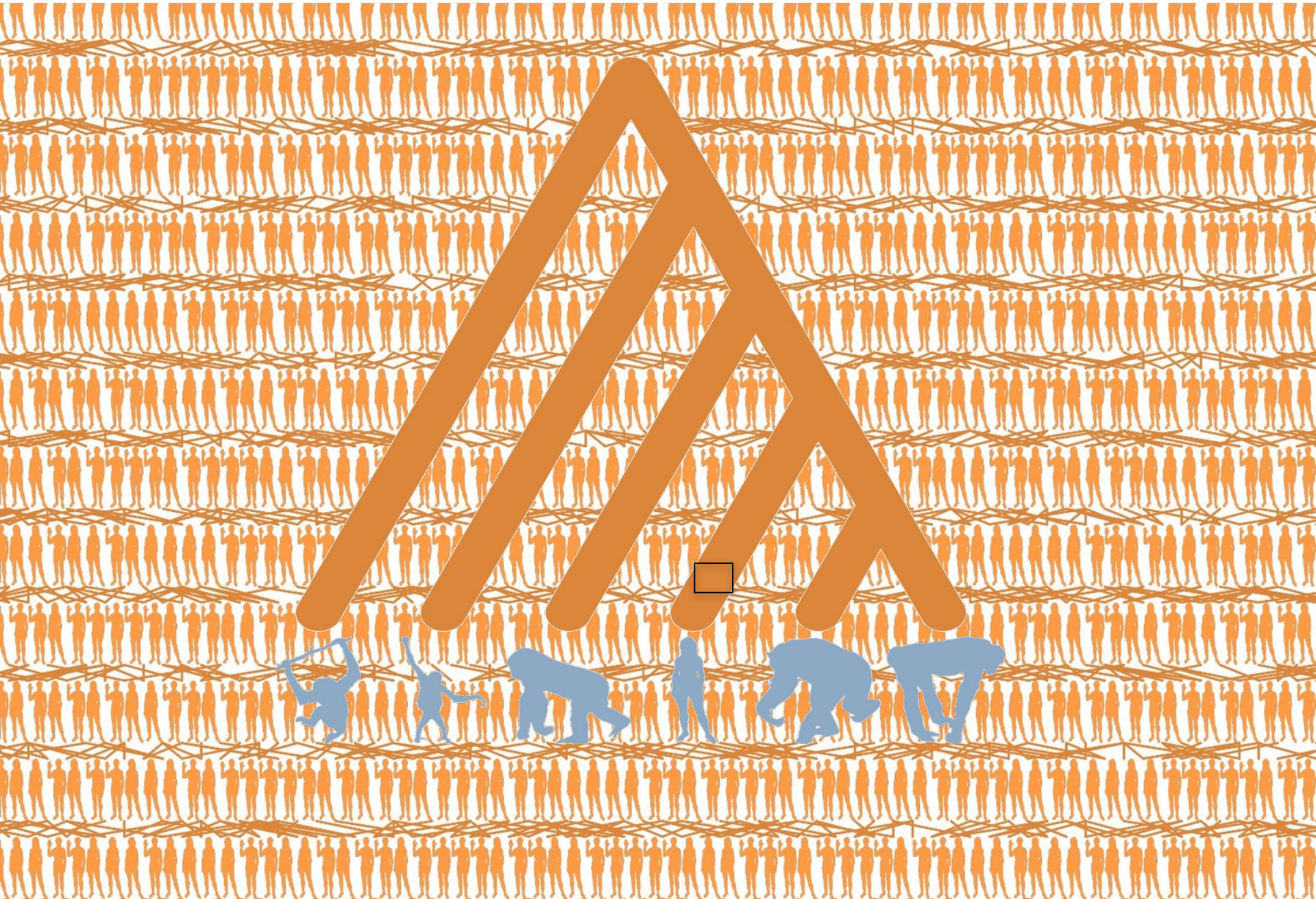
40 generations ago, you
would have > 1 trillion
ancestors

....2500 times the
population 800 years ago

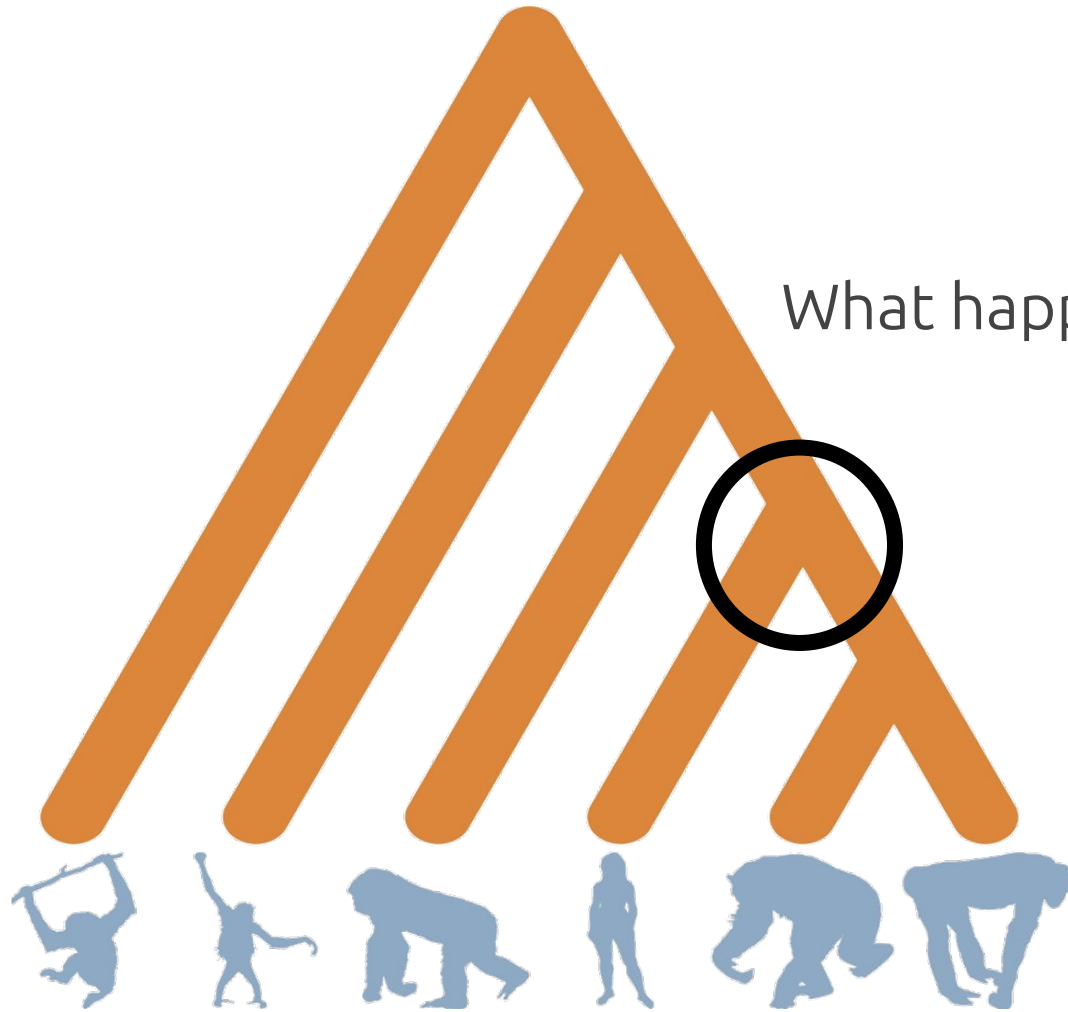
What does a phylogeny represent?



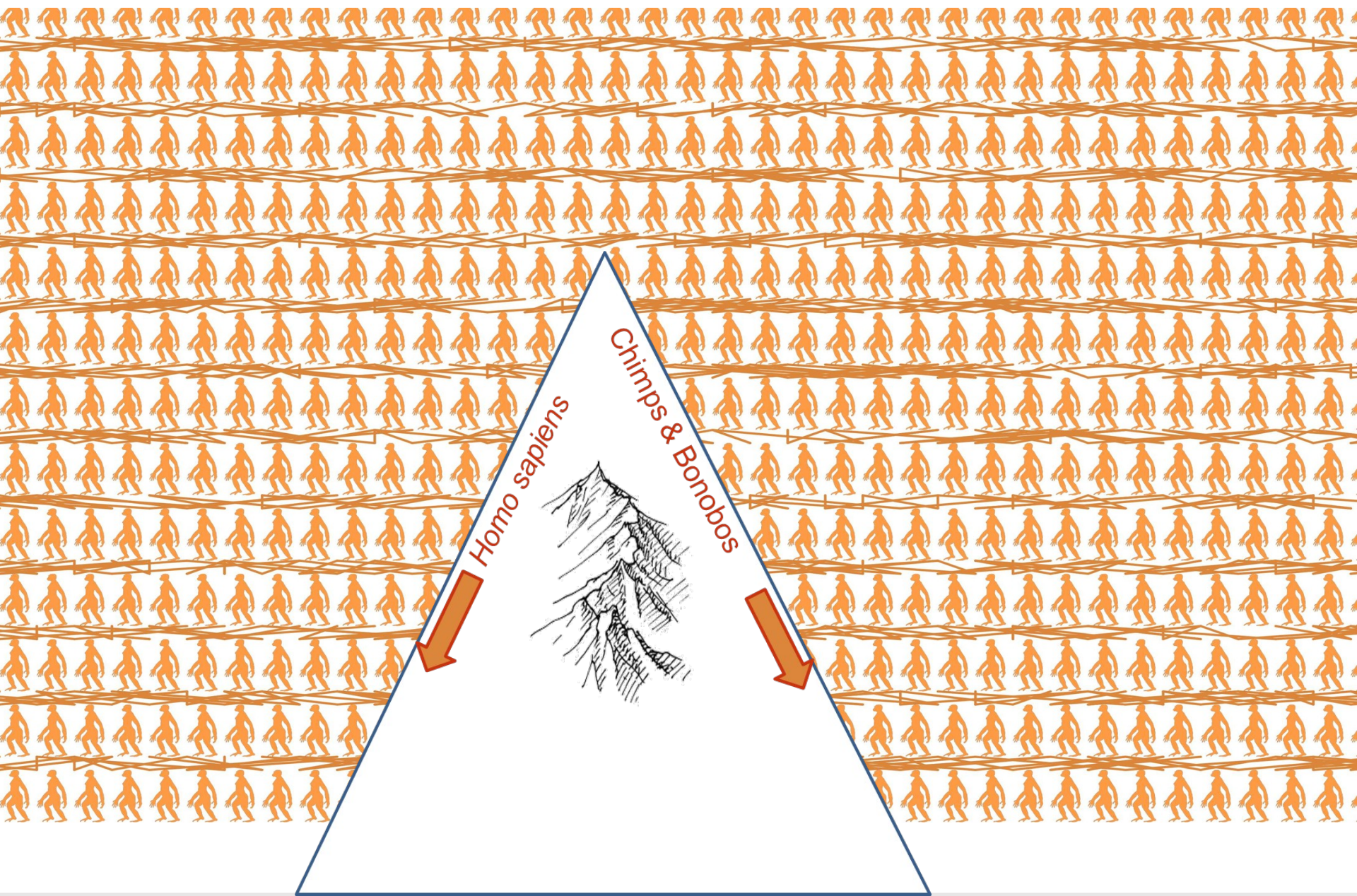
Branches represent population ancestry



What do nodes represent?



Nodes represent speciation events



Terminology

Root

Branch

Node

Tip

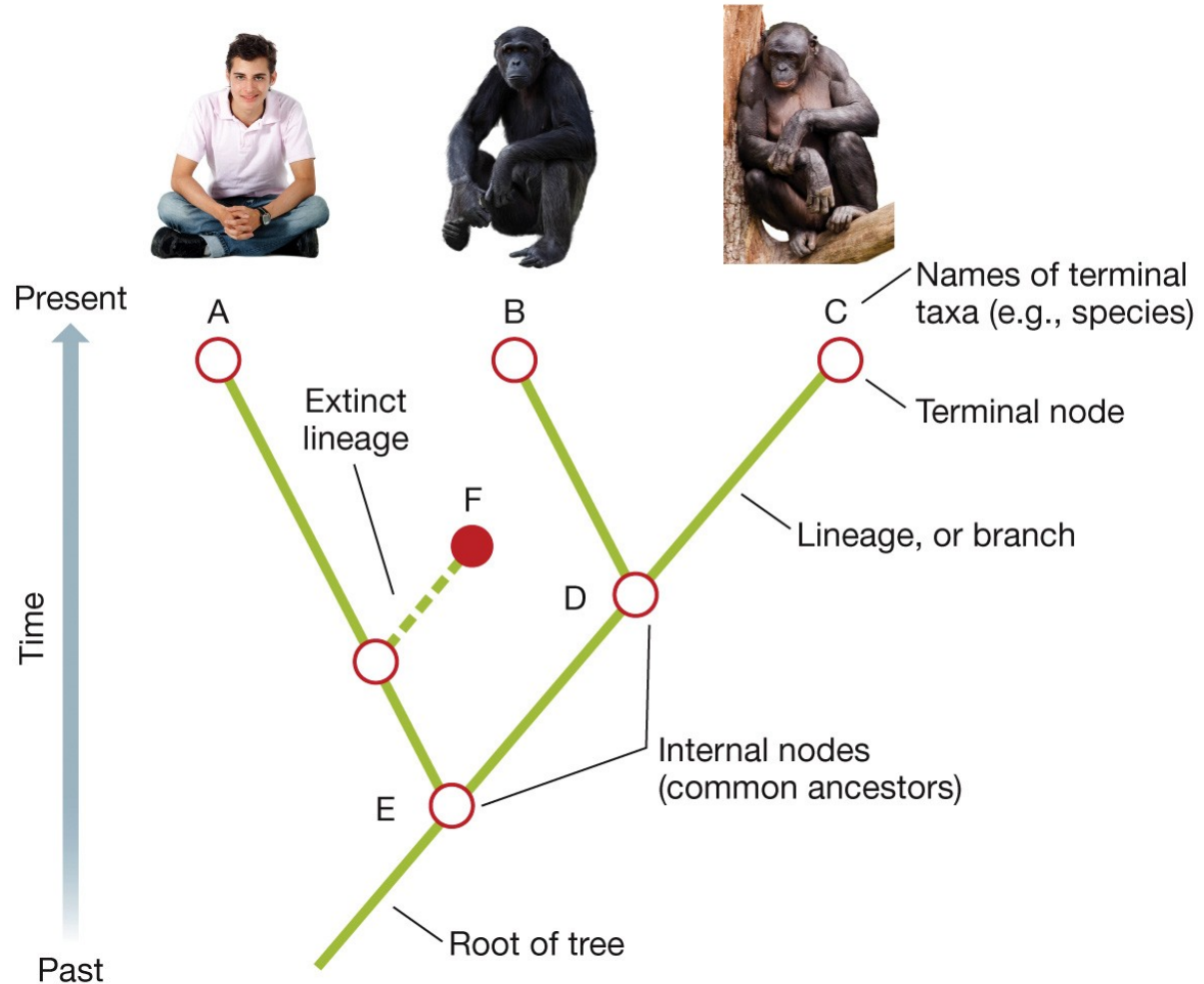
Outgroup

Clade

Monophyletic

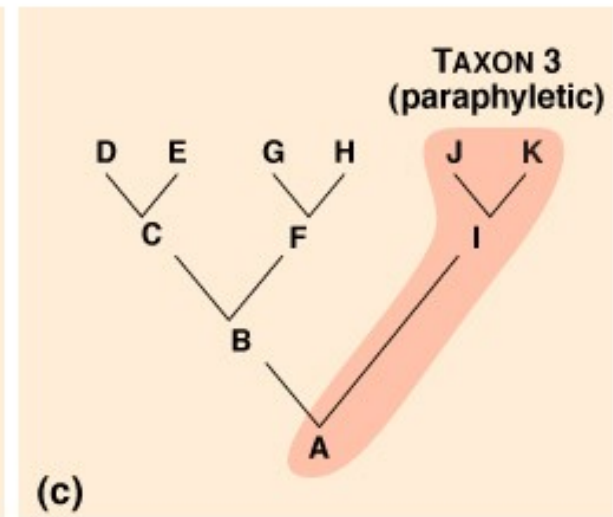
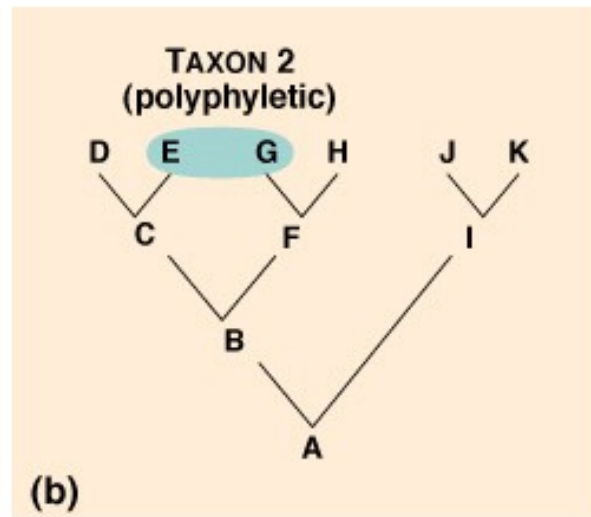
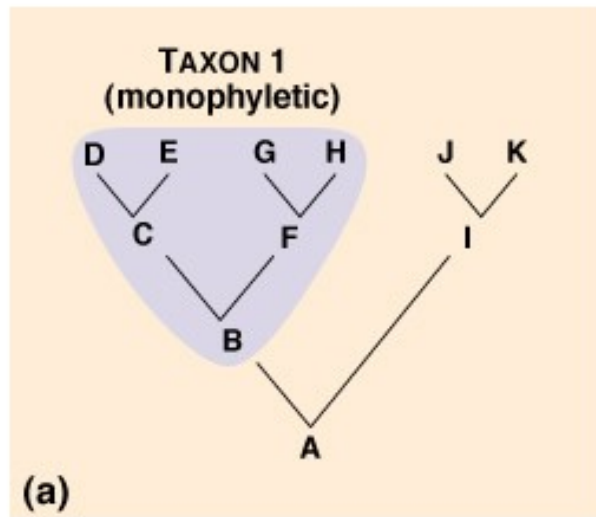
Paraphyletic

Polyphyletic



EVOLUTION 4e, Figure 2.6
© 2017 Sinauer Associates, Inc.

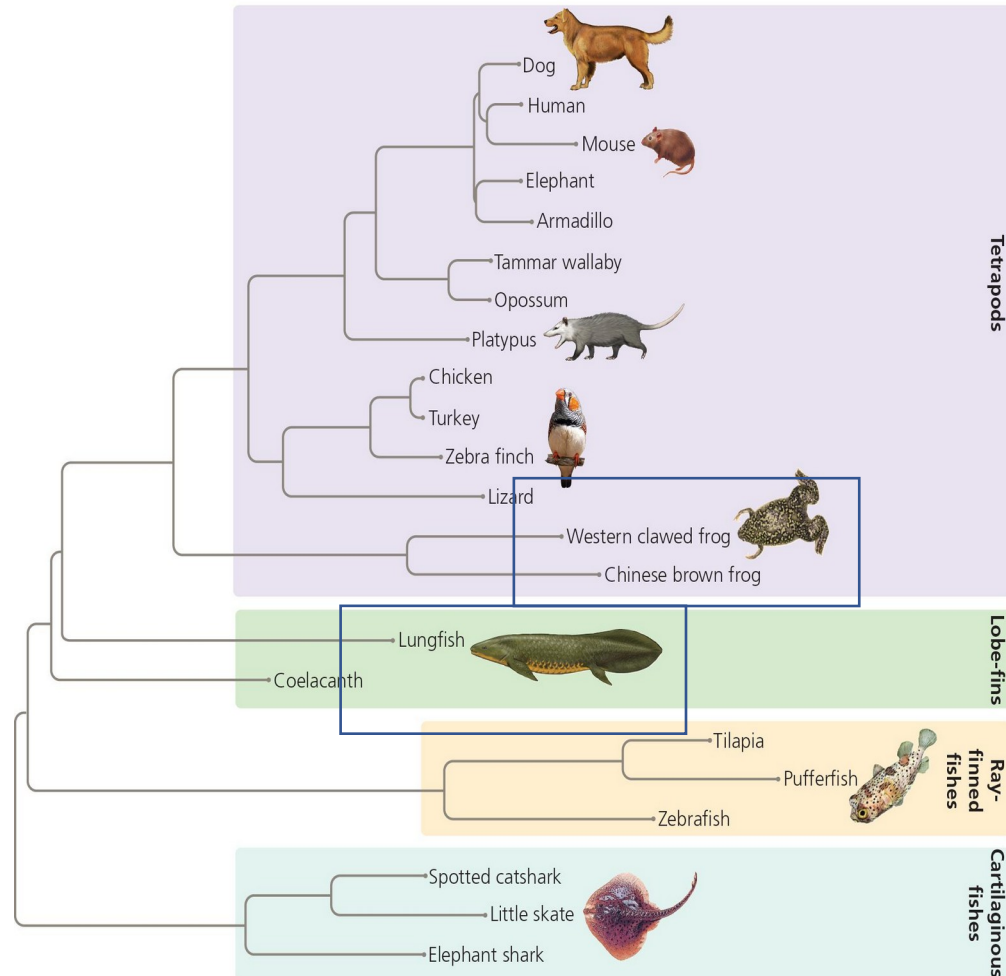
The goal of taxonomy and systematics is to represent monophyly



Practice question 1

The phylogeny depicted demonstrates that lungfish are basal to and more primitive than frogs.

- A. True
- B. False



Assignment I: Canvas self-assessment

Read Baum & Smith, and complete the questions in Assignment I. Take the canvas quiz. Retake up to 10 times until you get 100%. The purpose is to assess your understanding.

I'm going to assume you understand basic terminology and can read & interpret trees going forward. If you find it challenging (almost everyone does when first interpreting trees), come to office hour.

Macroevolution, systematics, phylogenetics, taxonomy, biodiversity science, phylogenetic comparative methods, oh my

What is macroevolution?

We seek to understand the processes that explain biodiversity.

My terminology for macroevolution

Inferring trees from data = Phylogenetics

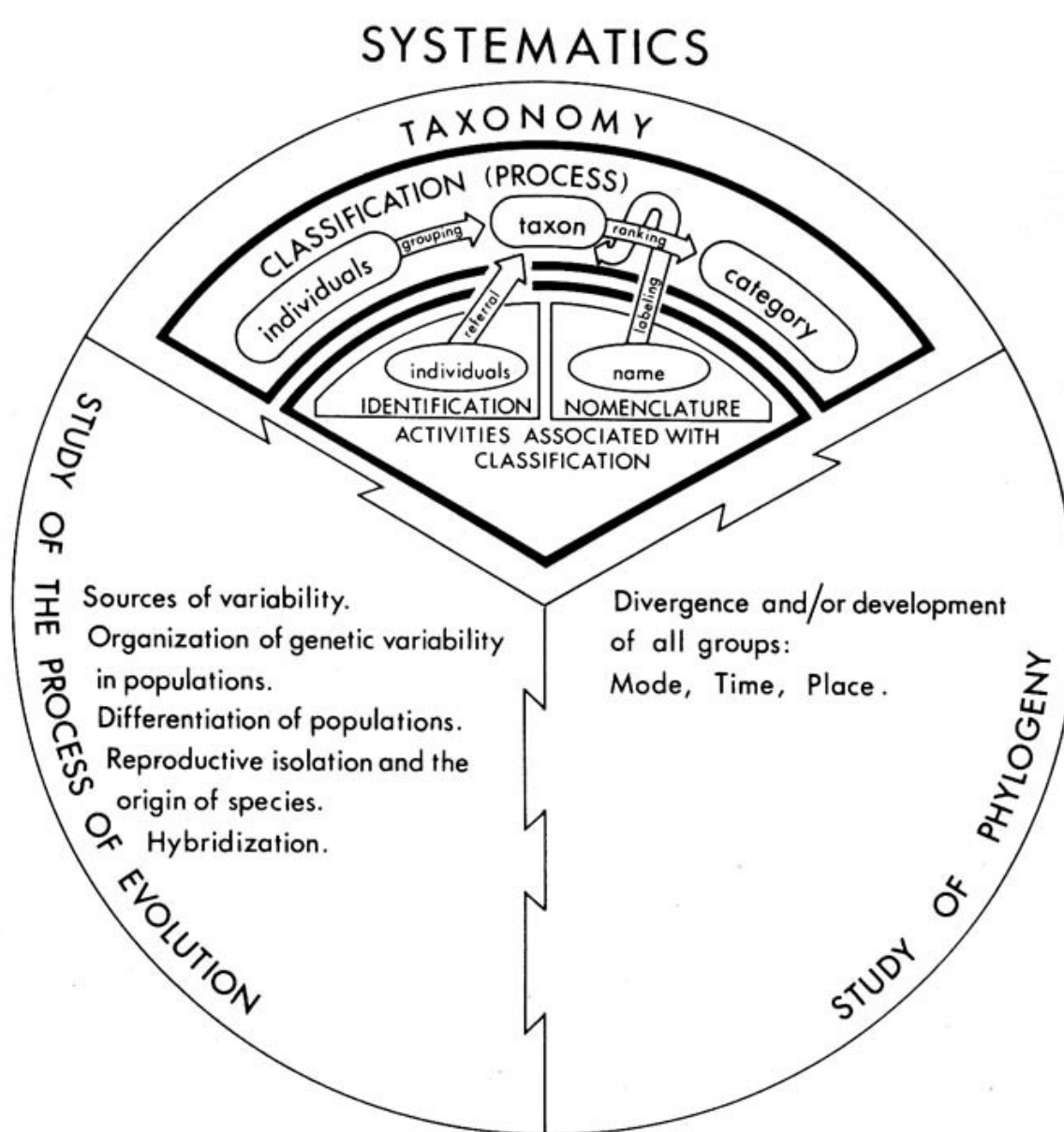
Using trees to study evolution = Phylogenetic comparative methods sensu lato

Asking evolutionary questions over deep time =
Macroevolution

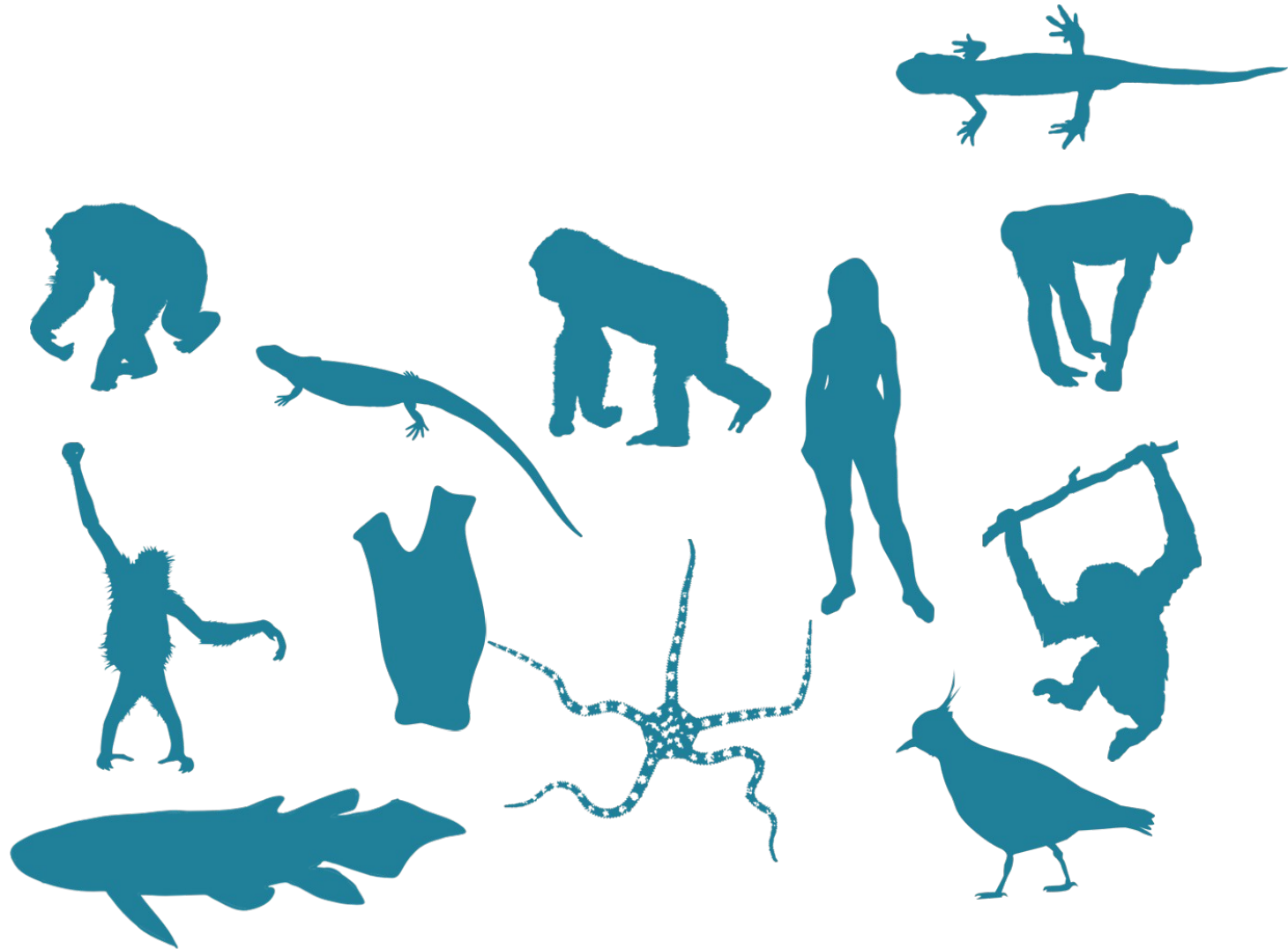
What is systematics?



Society of
Systematic Biologists



Going backwards- inferring phylogeny



Tree space is HUGE.

Table 3.1: The number of rooted, bifurcating, labeled trees for n species, for various values of n . The numbers for more than 20 species are approximate.

Inferring phylogenies is incredibly challenging statistical and computational task.

For that reason, phylogenetics is/has been at the forefront of many advances in statistics & computer science.

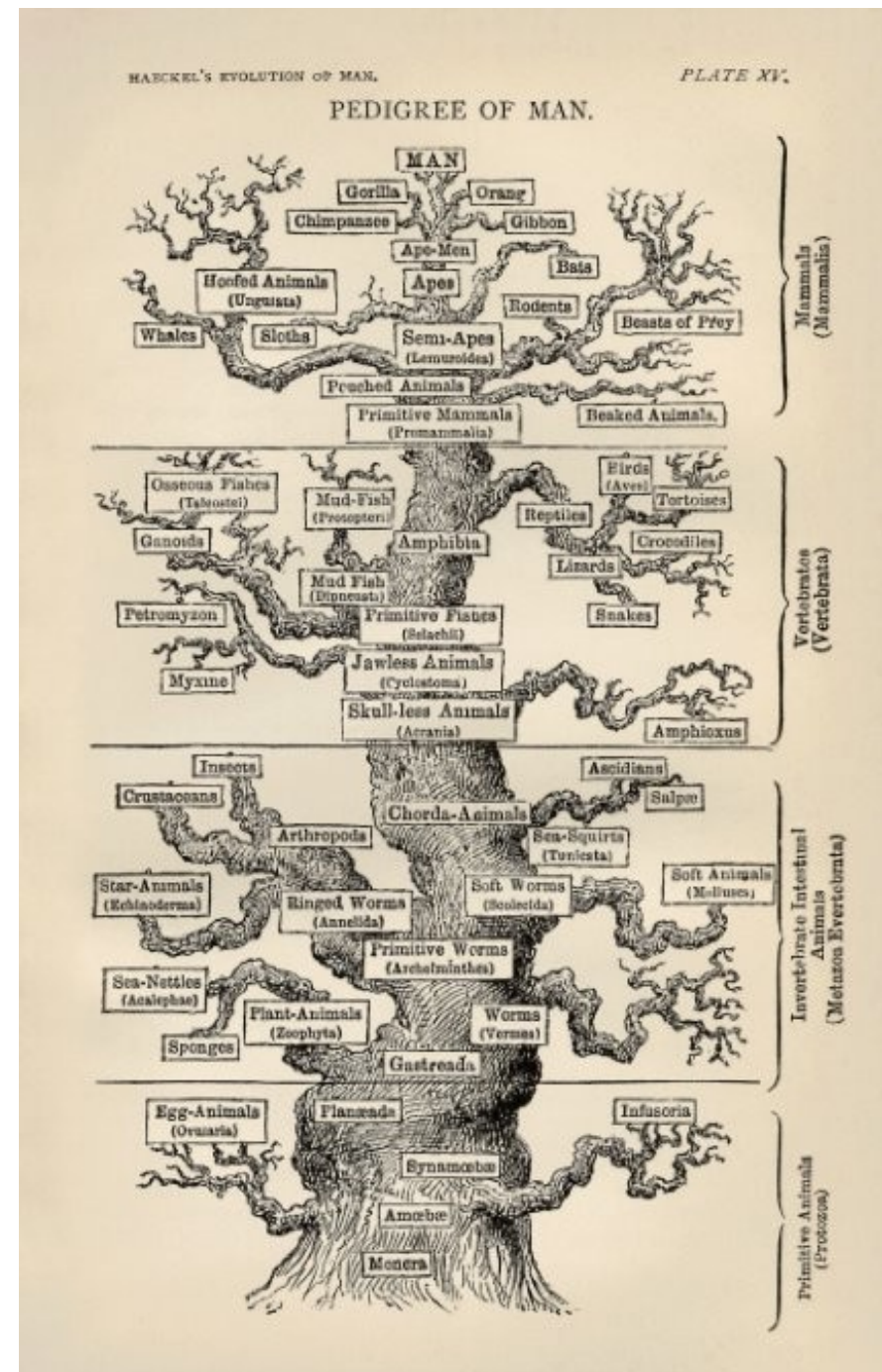
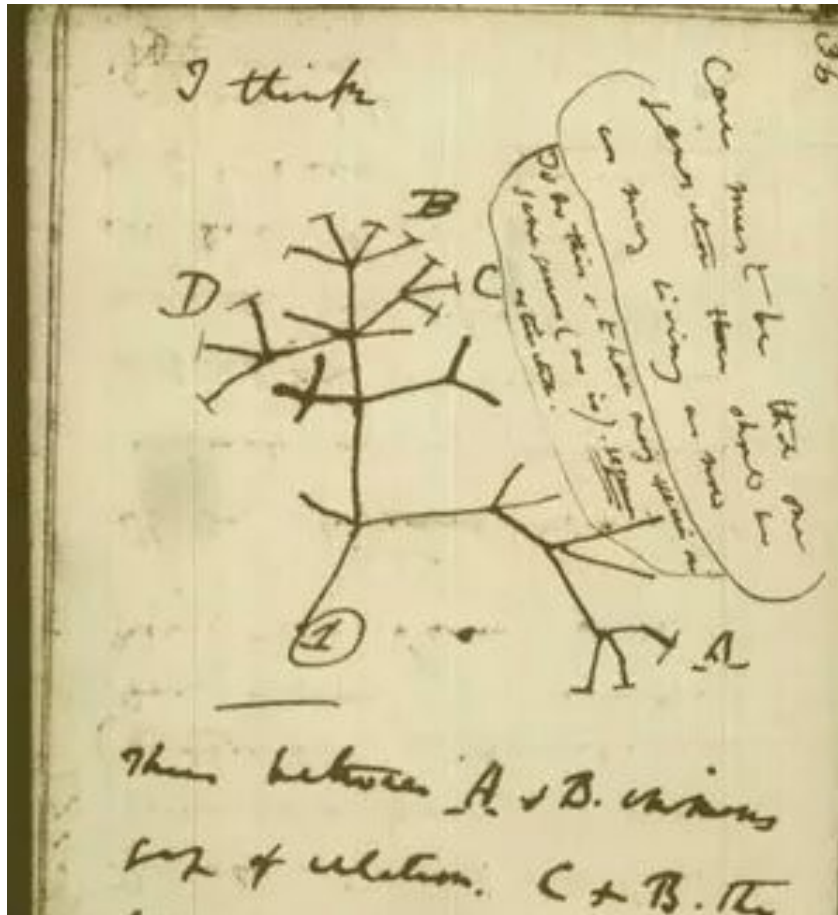
This course will cover the story of phylogenetics and the basic models, as they are shared with phylogenetic comparative methods.

For in depth “how-to” on building trees from molecular data, take Anton Suvorov’s course!

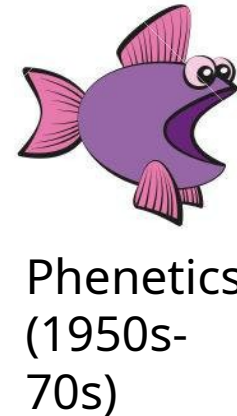
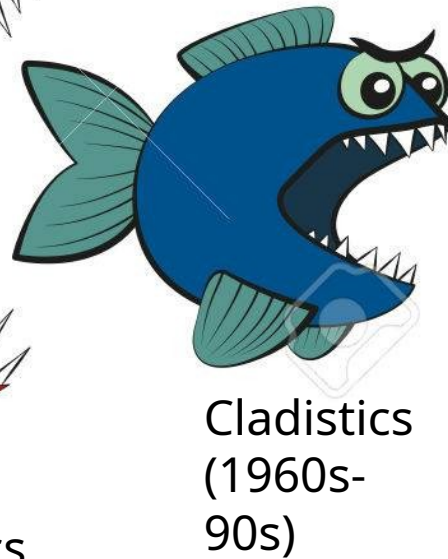
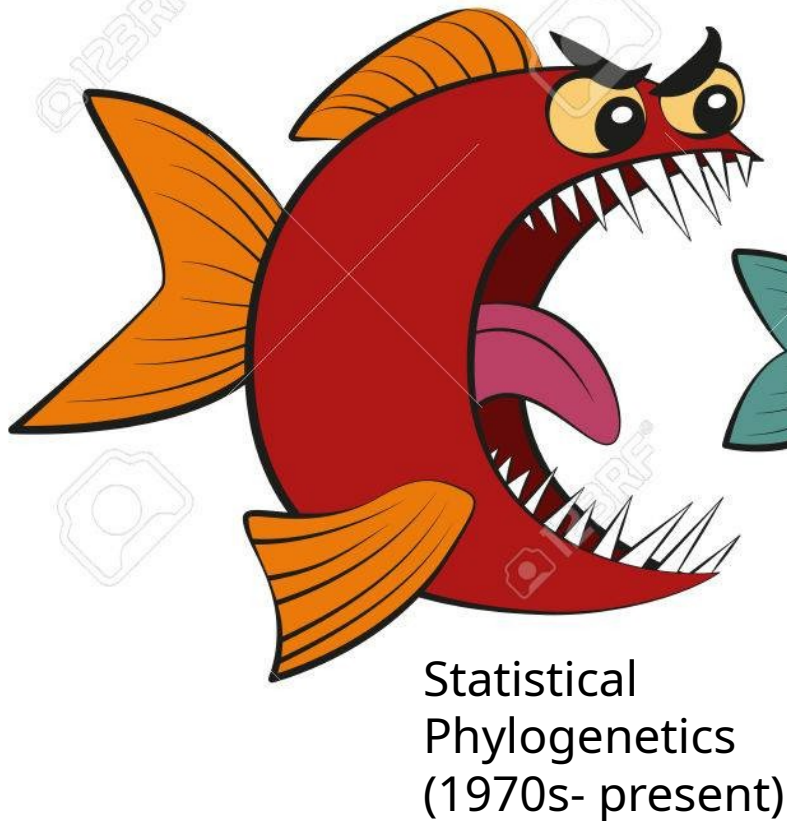
Species	Number of trees
1	1
2	1
3	3
4	15
5	105
6	945
7	10,395
8	135,135
9	2,027,025
10	34,459,425
11	654,729,075
12	13,749,310,575
13	316,234,143,225
14	7,905,853,580,625
15	213,458,046,676,875
16	6,190,283,353,629,375
17	191,898,783,962,510,625
18	6,332,659,870,762,850,625
19	221,643,095,476,699,771,875
20	8,200,794,532,637,891,559,375
30	4.9518×10^{38}
40	1.00985×10^{57}
50	2.75292×10^{76}

How can we find the "true" tree?
(Is there a true tree?!?)

A brief history of phylogenetics



The four cannibalistic schools of phylogenetics history



History of phylogenetics: Evolutionary taxonomy

No objective methodology

Relies on expert opinion
and a lifetime of study

Strengths? Weaknesses?



History of phylogenetics: Phenetics

Numerical Taxonomy & distance methods -
measure overall similarity

How do you measure similarity?

What are the drawbacks of this approach?

Sokal, Sneath & Rohlf (1960's) argued
phenetics provided much needed
reproducibility

History of phylogenetics: Phenetics

Numerical taxonomy and distance methods

Measure overall similarity

Sokal, Sneath, and Rohlf (1960s) argued
phenetics provided much needed reproducibility

How do you measure similarity?

What are the drawbacks of this approach?

Distance methods

Still used for fast tree construction

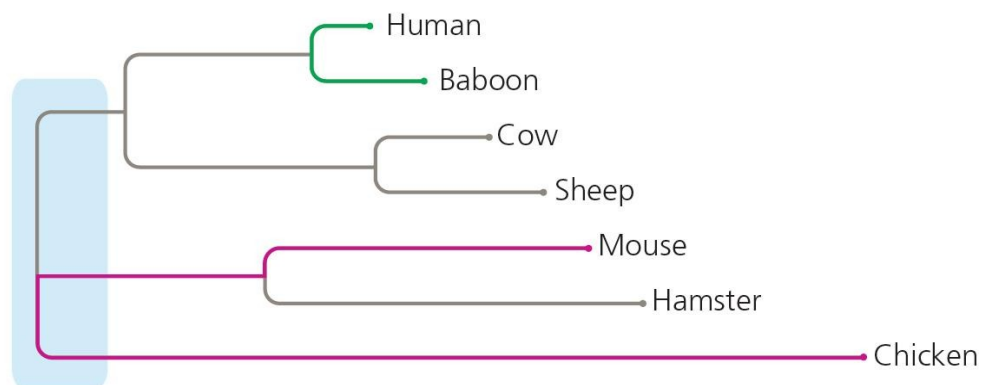
A

[illegible]**B**

B

	Baboon	Cow	Sheep	Mouse	Hamster	Chicken
Human	2	6	9	8	9	13
Baboon		7	10	7	10	13
Cow			3	11	12	16
Sheep				12	9	15
Mouse					7	16
Hamster						14

C



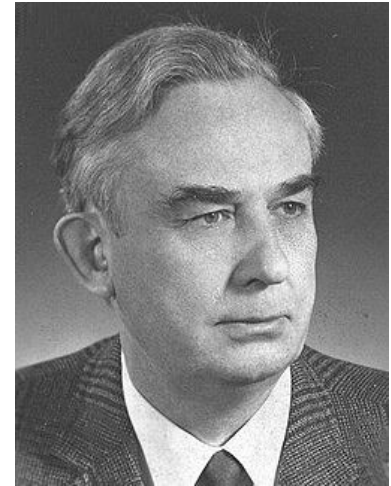
History of phylogenetics: Cladistics

Coined "phylogenetic systematics"

Grouping should not be based on *overall* trait similarity, but only on *shared derived traits* (*synapomorphies*)

Taxonomy based on *monophyly*

Parsimony & Occam's Razor- The simplest explanation is probably the correct one



Willi Hennig (1966)